24CSA111 - DATABASE MANAGEMENT SYSTEM (DBMS)

DBMS – Unit 3

- Normalization
- Anomalies
- Functional Dependency
- Armstrong's axioms
- Closure of a relation and Closure of attribute
- Lossless Decomposition
- 1NF, 2NF, 3NF, Boyce-Codd Normal Form

Normalization

Normalization is a process that helps analysis or database designers to design table structures for an application.

The focus of normalization is to attempt to reduce redundant table data to the very minimum.

Through this process, the collection of data in a single table is replaced, by the same data being distributed over multiple tables with a specific relationship being setup between the tables.

Anomalies

Anomaly is a problem that can occur in poorly planned un-normalized databases where all the data is stored in one table.

Types of Anomalies:

- Insert – it occurs when certain attributes cannot be inserted into the database without the presence of other attributes.
- Delete – it exists when certain attributes are lost because of the deletion of other attributes.
- Update – it exists when one or more instances of duplicated data is updated, but not all.

Functional Dependency

A functional dependency [FD or f.d.], denoted by X->Y, between two sets of attributes X and Y that are subsets of R specifies a constraint on the possible tuples that can form a relation state r of R.

The constraint is that, for any two tuples t1 and t2 in r that have t1[X]=t2[X], they must also have t1[Y]=t2[Y].

Armstrong's Axioms

- Armstrong's Axioms is a set of rules.
- It provides a simple technique for reasoning about functional dependencies.
- It was developed by William W Armstrong in 1974.
- It is used to infer all the functional dependencies on a relational database.

Armstrong's Axioms – Rules

- Reflexivity – If A is a set of attributes and B is a subset of A, then A holds B. {A -> B}
- Augmentation – If A hold B and C is a set of attributes, then AC holds BC. {AC -> BC}
- Transitivity – If A holds B and B holds C, then A holds C.
  - If {A->B} and {B->C} then {A->C}

Closure of a relation and Closure of attribute

- The set of all those attributes which can be functionally determined from an attribute set is called as a closure of that attribute set.
- Closure of attribute set {X} is denoted as $\{X\}^+$.

- Given a relation R on a set A and a property P of relations, the closure of R with respect to property P, denoted by Clp(R), is smallest relation on A that contains R and has property P.

Lossless Decomposition

- Decomposition is lossless if it is feasible to reconstruct relation R from decomposed tables using joins.
- The information will not lose from the relation when decomposed.
- The join would result in the same original relation.

1NF

The First Normal Form (1NF), states that the domain of an attribute must include only atomic values and that the value of any attribute in a tuple must be a single value from the domain of that attribute.

Un-Normalized Data:

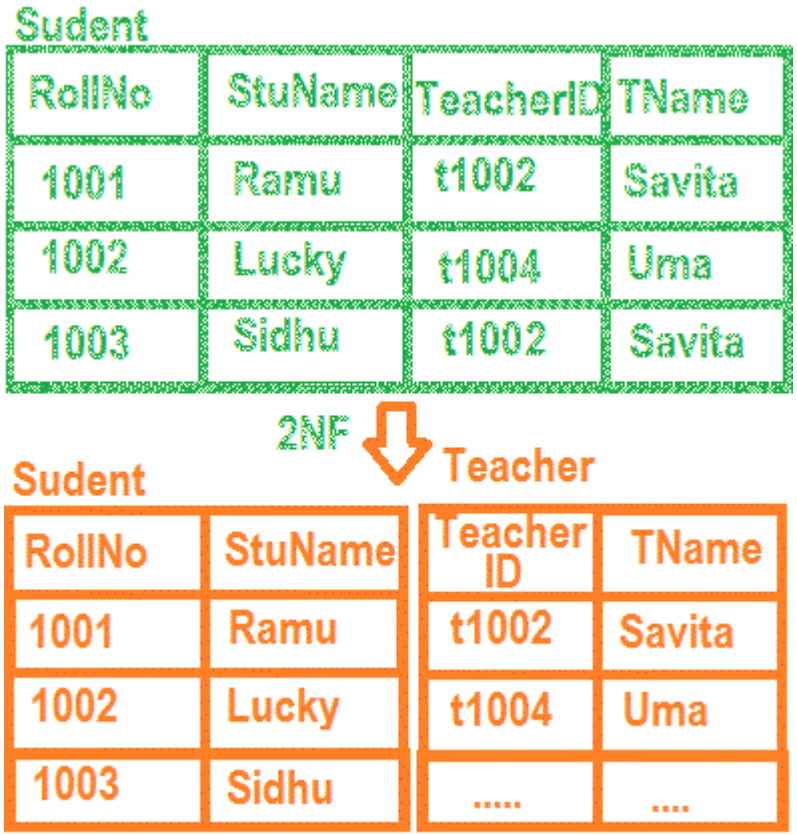
Manager	Persons
Vinod	Shaju , Manoj, Ashok, Naveen
Rajiv	Saravana, Salil, Kannan

1NF

Manager	Persons
Vinod	Shaju
Vinod	Manoj
Vinod	Ashok
Vinod	Naveen
Rajiv	Kannan
Rajiv	Salil
Rajiv	Saravana

2NF

A relation schema R is in 2NF, if every nonprime attribute A in R is fully functionally dependent on the primary key of R.



3NF

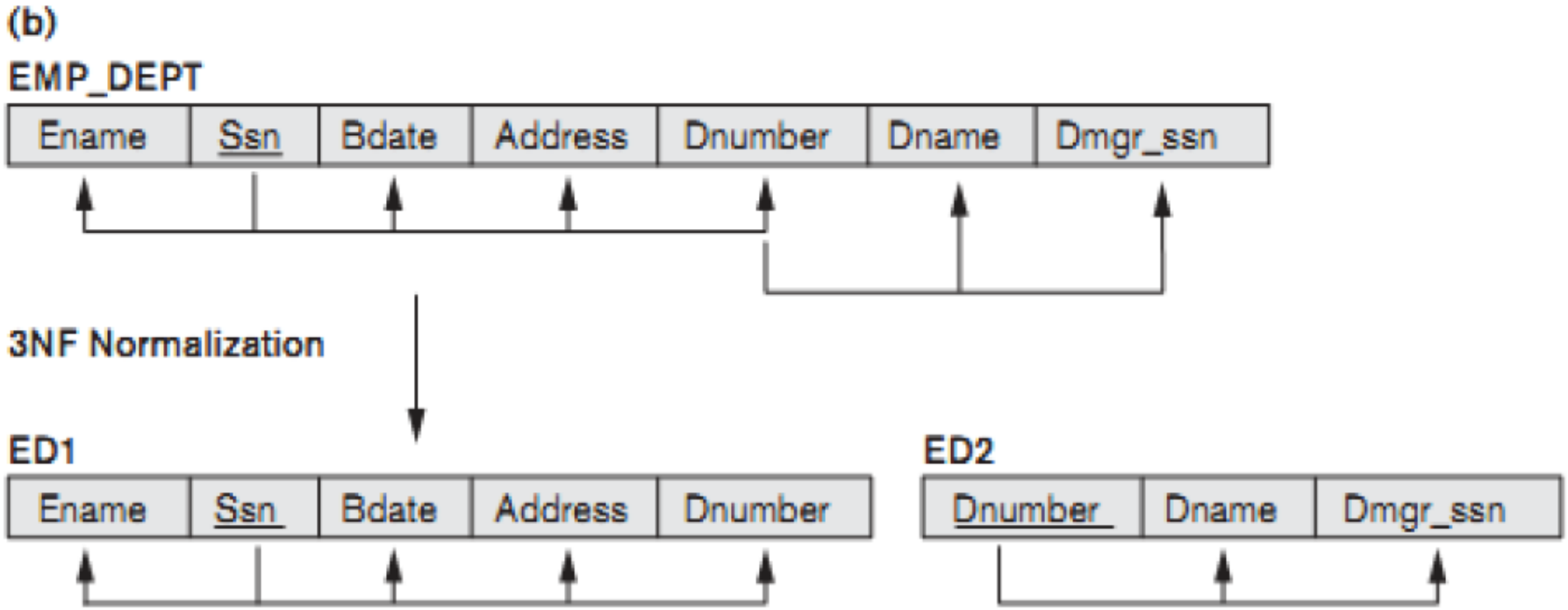
A relation schema R is in 3NF, if it satisfies 2NF and no nonprime attribute of R is transitively dependent on the primary key. The 3NF is based on the concept of Transitive Dependency.

A general case of transitive dependencies is as follows:

A, B, C are three columns in a table.

If C is related to B and If B related A, then C is indirectly related to A. This is when **transitive dependency** exists.

3NF



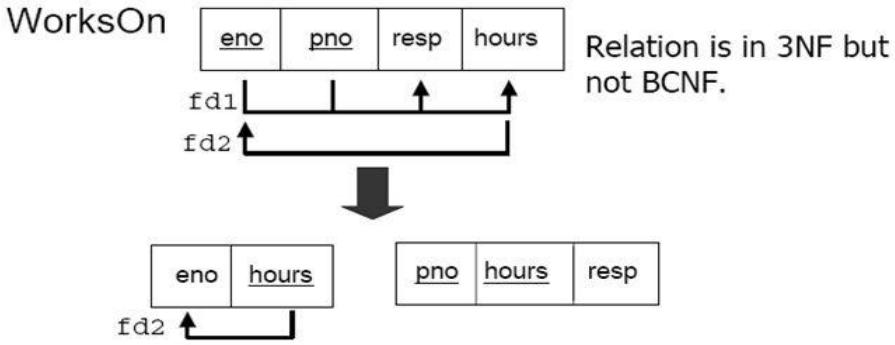
The dependency Ssn->Dmgr_ssn is transitive through Dnumber in EMP_DEPT, because both the dependencies Ssn->Dnumber and Dnumber->Dmgr_ssn hold and Dnumber is neither a key itself nor a subset of the key of EMP_DEPT. (Dnumber is not a key of EMP_DEPT)

Boyce-Codd Normal Form

A relation schema R is in BCNF if whenever a nontrivial functional dependency X->A holds in R, then X is a super key of R.

Boyce-Codd Normal Form (BCNF)

- Consider the WorksOn relation where we have the added constraint that given the hours worked, we know exactly the employee who performed the work. (i.e. each employee is FD from the hours that they work on projects). Then:



Note that we lose the FD eno,pno → resp, hours.

THANK YOU